

KINETICS OF FLOW AND STRAIN-HARDENING†

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Abstract—The kinetics of glide at constant structure and the kinetics of structure evolution are correlated on the basis of various experimental observations in pure f.c.c. mono- and polycrystals. Two regimes of behavior are identified. In the initial regime, the Cottrell–Stokes law is satisfied, hardening is athermal, and a single structure parameter is adequate. With increasing importance of dynamic recovery, be it at large strains or at high temperatures, all of these simple assumptions break down. However, the proportionality between the flow stress and the square-root of the dislocation density holds, to a good approximation, over the entire regime; mild deviations are primarily ascribed to differences between the various experimental techniques used. A phenomenological model is proposed, which incorporates the rate of dynamic recovery into the flow kinetics. It has been successful in matching many experimental data quantitatively.

Résumé—Nous avons relié la cinétique du glissement à structure constante et de l'évolution de la structure à partir de diverses observations sur des monocristaux et des polycristaux c.f.c. purs. Nous avons distingué deux régimes. Au cours du premier régime, la loi de Cottrell et Stokes est vérifiée, le durcissement est athermique et il suffit d'un seul paramètre pour la structure. Lorsque l'importance de la restauration dynamique augmente, que ce soit aux grandes déformations ou aux températures élevées, toutes ces hypothèses simples s'effondrent. Cependant, la contrainte d'écoulement reste proportionnelle à la racine carrée de la densité de dislocations, à une bonne approximation, dans tout le régime; quelques faibles déviations sont principalement attribuées aux différences entre les diverses techniques expérimentales utilisées. Nous proposons un modèle phénoménologique, qui permet d'introduire la vitesse de la restauration dynamique dans la cinétique de l'écoulement. Il a permis de rendre compte quantitativement d'un certain nombre de résultats expérimentaux.

Zusammenfassung—Anhand verschiedener experimenteller Beobachtungen an reinen kfz. Ein- und Vielkristallen wurden die Kinetik der Gleitung bei konstanter Struktur und die Kinetik der Strukturentwicklung miteinander korreliert. Zwei Bereiche unterschiedlichen Verhaltens wurden aufgefunden. Im Anfangsbereich ist das Cottrell–Stokes-Gesetz erfüllt, die Verfestigung ist athermisch und ein einziger Strukturparameter ist ausreichend. Mit zunehmendem Einfluß der dynamischen Erholung bei großer Dehnung oder hoher Temperatur werden alle diese einfachen Annahmen ungültig. Der lineare Zusammenhang zwischen Fließspannung und Wurzel aus der Versetzungsdichte dagegen bleibt in guter Näherung im gesamten Bereich bestehen. Kleinere Abweichungen werden im wesentlichen den Unterschieden zwischen den verschiedenen experimentellen Techniken zugeschrieben. Es wird ein phänomenologisches Modell vorgeschlagen, welches die Geschwindigkeit der dynamischen Erholung bei der Kinetik des Fließens berücksichtigt. Dieses Modell kann viele experimentelle Daten quantitativ beschreiben.

INTRODUCTION: STRUCTURE AND CHANGE

It is a fundamental tenet of materials science that the current properties of a piece of material depend entirely on its current metallurgical structure, and that changes in properties result from changes in structure. The structure may, however, change as the property is being tested, for example by aging. The dislocation structure occupies a special place in its relation to mechanical properties, because it necessarily changes during the test. This is obviously true in a dynamic sense, since it is dislocation motion that causes the

deformation; but it is also true in a permanent sense when the material strain hardens.

A classical abstraction that divides plastic behavior into two mechanistic steps has nevertheless proved useful: the flow stress σ depends on the *current* structure

$$\sigma = \sigma(\rho; \dot{\epsilon}, T) \quad (1a)$$

and the structure evolves with strain ϵ

$$\frac{d\rho}{d\epsilon} = E(\rho). \quad (1b)$$

In these simplified relations, it has been assumed that the dislocation structure can be represented by the single parameter ρ , the 'dislocation density' (i.e., some appropriate average of the dislocation distribution). We have further assumed that, at given ρ , the flow

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stress depends on strain rate $\dot{\epsilon}$ and temperature T , but that the rate of evolution of ρ does not. Such a rate-independence of strain hardening is usually implied in phenomenological plasticity. More general cases, and more specific ones, will be discussed below.

A measurement of the flow stress 'at given structure' [equation (1a)] requires that a suitable back-extrapolation method has been found which eliminates all effects of ongoing structure change [1, 2]. This is not necessarily possible, even in principle: the flow stress may also depend on the current *rate of change* of the dislocation structure [3–5], so that the separation of constant-structure kinetics from evolution is not as clear-cut as was implied in the relations (1a) and (1b) [2]. It is the purpose of this paper to present and review a number of experimental observations that are most easily explained by such an assumption, and are incompatible with relations (1). The same experiments show, on the other hand, that the simple relations (1) are in fact well obeyed over a certain range of external conditions: generally speaking, low temperatures and low strains.

First, let us generalize the formal relations (1) in two respects: allowing that not only the flow stress, but also the rate of evolution of the substructure with strain may depend on strain rate and temperature; and that it requires *two* parameters (here called ρ_1 and ρ_2) to represent the macroscopic effects of the substructure adequately:

$$\sigma = \sigma(\rho_1, \rho_2; \dot{\epsilon}, T) \quad (2a)$$

Then, *two* evolution laws are required:

$$\frac{d\rho_1}{d\epsilon} = E_1(\rho_1, \rho_2; \dot{\epsilon}, T) \quad (2b)$$

$$\frac{d\rho_2}{d\epsilon} = E_2(\rho_1, \rho_2; \dot{\epsilon}, T) \quad (2c)$$

This set of relations will be sufficiently general for all applications to be discussed here. An extension to further parameters may be necessary when stress reversals are incorporated [6].

Comparison of relations (2) and (1) reveals a significant difference relevant to phenomenological descriptions of plasticity, in which the metallurgical *structure* is of no concern, but instead the notion of a mechanical *state* of the material is introduced. When relation (1a) holds, a measurement of the flow stress at any standard strain rate and temperature identifies the current state of the material [7–9]: it is a macroscopic equivalent to ρ . The state of a material obeying equation (2a), on the other hand, cannot be completely specified by one measurement [9], such as of the flow stress at standard conditions (the 'reference stress', the

'plastic resistance'); however, a measurement of both the current flow stress and the current strain-hardening rate might be sufficient. It is seen that the above-mentioned possibility of a dependence of the flow stress on the current rate of evolution could then be a consequence of its dependence on *two* structure parameters. We will find such a correlation borne out in the experiments.

FLOW STRESS AND DISLOCATION DENSITY

As a point of departure, we will discuss the specific relation between flow stress and dislocation density that is in common usage. The flow stress 'at zero temperature' is set equal to

$$\hat{\sigma} = \hat{\alpha}\mu b\sqrt{\rho} \quad (3)$$

where b is the magnitude of the Burgers vector, μ an appropriate [7] shear modulus and $\hat{\alpha}$ a constant of order unity which depends, in part, on the strength of the dislocation/dislocation interaction. Thermal activation may lower this effective obstacle strength [7] so that the flow stress at a finite temperature and strain rate becomes

$$\sigma = s(\dot{\epsilon}, T) \cdot \hat{\alpha}\mu b\sqrt{\rho} \equiv \alpha\mu b\sqrt{\rho} \quad (4)$$

where $s(\dot{\epsilon}, T)$ is a function that goes to 1 as $T \rightarrow 0$; it will be discussed later, on the basis of thermal activation kinetics. For now, it matters only that the flow stress is a *product* of a rate sensitive† term and a structure sensitive term.

The flow stress as given by equation (3) and (4) relates only to the impediment to dislocation motion that is provided by other dislocations. In most materials, there are other contributions to the plastic resistance. In some cases (e.g., lattice resistance [7], solution hardening [7], some grain size effects [10]), these are additive to the contributions discussed above:

$$\sigma = \alpha(\dot{\epsilon}, T) \cdot \mu b\sqrt{\rho} + \sigma_0(\dot{\epsilon}, T). \quad (5)$$

The rate dependence of σ_0 may be more important than that of α (or s), or it may be negligible; the less rate sensitive term is often called an 'internal stress'. Some other contributions to the flow stress obey more complicated, nonlinear superposition rules [11]. For the purposes of this paper, we assume that the superposition problem has been solved so that, from now on, σ identifies *only the dislocation/dislocation interaction component of the flow stress, which changes with strain* (equation 4). The experimental results we will discuss relate only to pure f.c.c. materials, in which all other contributions can be shown to be negligible (except at very small stresses). Furthermore, we presuppose that self-consistent use of orientation factors is made in any application; we use σ and ϵ in the general sense of 'equivalent' stresses and strains, and use τ and γ only when experimental data are specifically reported in terms of resolved quantities.

† All rate and temperature dependencies of interest here are due to thermal activation and are thus coupled. When the influence of thermal activation is important, we say the behavior is 'rate sensitive'; when not, 'athermal'. The temperature dependence of μ is assumed known and incorporated in the analysis.

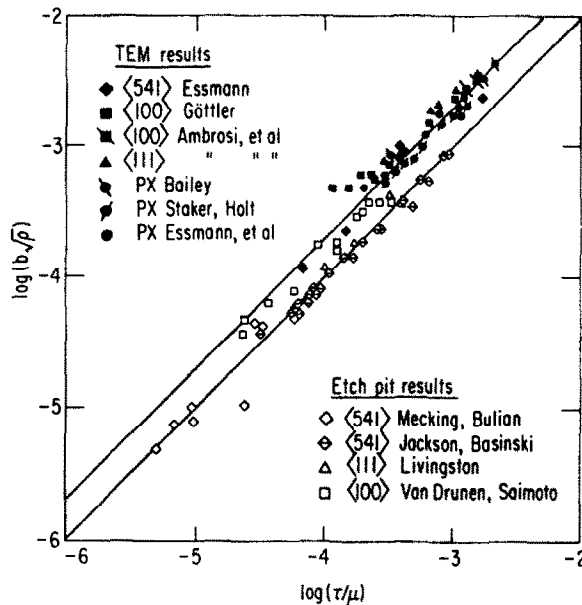


Fig. 1. Dislocation density vs flow stress for Cu at room temperature [12–21], normalized by $b = 0.256$ nm, $\mu = 42.1$ GPa (decadic logarithms). Polycrystal (PX) tensile stresses were divided by 3.06 to convert to shear stress τ . Volume dislocation densities measured by TEM were divided by 2 to convert to intersection density ρ . For single slip $\langle 541 \rangle$, etch-pit data refer to forest density, TEM data to dislocations with nonprimary Burgers vectors. From the low stresses, 0.12 MPa was subtracted [20]. The lines show the relation $\sigma = \alpha \mu b \sqrt{\rho}$, with $\alpha = 1$ and 0.5 .

The relation between flow stress and dislocation density has been investigated over the widest range of the variables, and under the widest range of experimental conditions, in pure copper. A large variety of these data [12–21] has been replotted in Fig. 1 in a way that would make departures from the square-root relation [equations (4)] most clearly visible. All of these data refer to room-temperature deformation,† all are expressed in terms of resolved shear stress and normalized by the same shear modulus $\mu \equiv \sqrt{c_{44} \cdot (c_{11} - c_{12})/2}$, and all are reported as intersection (rather than volume) densities [12]. For the single-slip data, ρ is the forest density in the case of etch-pit studies [15†, 20]; in the one reported single-slip investigation by transmission electron microscopy (TEM) [14], ρ is the density of dislocations with nonprimary Burgers vectors, as measured after neutron irradiation. In the cases of polyslip deformation, ρ refers to the 'total' dislocation density [12, 13, 16–19]. The lowest stresses were corrected by a constant additive contribution τ_0 of 0.12 MPa, presumably due to impurities and chosen such as to produce a fit with equation (5) [20].

† The single most extensive investigation [22] was undertaken at $T = 4.2$ K. Even if the mechanisms were identical to those at R.T., a conversion of the data would require a knowledge of the temperature dependence of the impurity contribution. Furthermore, the published data contained an unknown, specimen-dependent yield stress parameter.

‡ For this reason, Livingston's [12] single-slip data are reported in the reevaluation by Jackson and Basinski [15].

The data in Fig. 1 span three orders of magnitude in stress, almost seven in dislocation density (from about 10^4 to 10^{11} cm $^{-2}$). Two observations must be made at first sight: each individual set of data obeys the square-root relation quite well; and all data together are compatible with it if a total variation in α of a factor 2 is allowed, i.e. a variation in dislocation density of a factor 2 either way from the mean. It would seem that such a variation between data from different investigators could easily be due to experimental scatter or differences in evaluation techniques.

Beyond this first-order evaluation, Fig. 1 may suggest a correlation between lower values of α and higher dislocation densities, or a systematic departure from the square-root relation [21–24]. We find it significant, however, that at least two other correlations can be made: TEM data (solid symbols) give consistently higher dislocation densities than etch-pit data (open symbols); and polyslip deformation (in mono- or polycrystals) gives consistently higher densities than single slip [21]. For both of these correlations, plausible explanations are at hand. We find the difference in measurement technique the most convincing argument, as well as the one with the best experimental correlation: all etch-pit data fit equation (5) with $\alpha \approx 1$, all TEM data with $\alpha \approx 0.5$. Note, on the other hand, that if the data were fit by one straight line of slope greater than one, the deviations are systematic, namely of a sigmoidal nature. This is also evident in the plot used by Basinski [22, 24].

In conclusion, the proportionality between the flow stress (due to dislocation/dislocation interactions) and

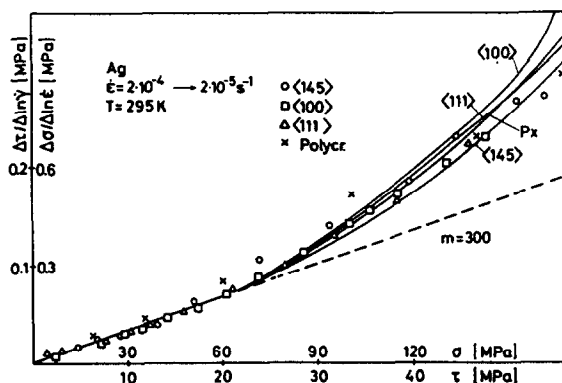


Fig. 2. The rate sensitivity of the flow stress vs the flow stress, both as a function of prestrain, for silver mono- and polycrystals [27]. The data points were determined from instantaneous stress jumps upon a decrease in strain rate. The solid lines here and in the next two figures are predictions from a model to be presented below, with one adjustable parameter listed in parenthesis.

the square-root of the (forest) dislocation density is well substantiated. Mild departures in the direction of less rapidly rising stress at higher densities are compatible with the experiments, but not clearly derivable from them.

FLOW-STRESS KINETICS

The rate sensitivity of the flow stress is most conveniently presented on a 'Haasen plot' [7, 11, 24, 25], namely as rate sensitivity β vs flow stress σ , both varying with strain. If equation (4) is valid, this plot should be a straight line through the origin:

$$\beta \equiv \left. \frac{\partial \sigma}{\partial \ln \dot{\epsilon}} \right|_{\rho, \tau} = \frac{\partial \ln s}{\partial \ln \dot{\epsilon}} \bigg|_{\tau} \cdot \sigma. \quad (6)$$

The straight-line relation follows from the assumed independence of s on ρ or, in other words, from the product nature of equation (4) [26]. If there were other, constant contributions σ_0 to the flow stress (equation 5), the line would remain straight but would no longer go through the origin [11].

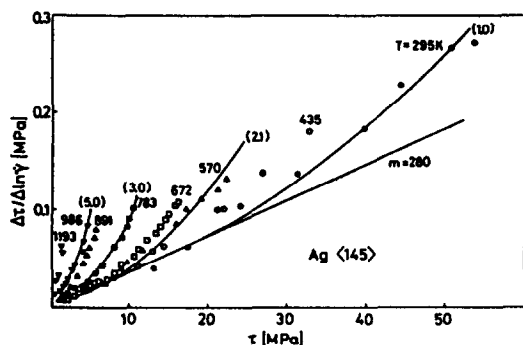


Fig. 3. As Fig. 2, only for single crystals in single slip, but for a wide range of test temperatures.

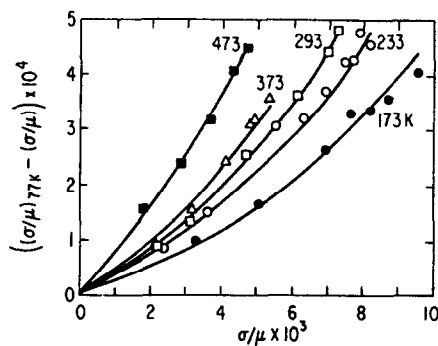


Fig. 4. Change in flow stress (normalized by temperature dependent shear modulus) upon a decrease in test temperature from the temperatures given to 77 K, vs tensile flow stress at temperature, in copper polycrystals, both as a function of strain. Data points evaluated from Bullen and Hutchison [29].

Figure 2 shows such a Haasen plot for silver single crystals in single slip and polyslip, and silver polycrystals [27]. The measurements refer to the maximum stress change in the initial transient after a decrease in strain rate by a factor 10. It is seen that all data points fall well on a straight line through the origin for a finite range of flow stresses, but deviate systematically above a fairly well defined value of the flow stress, which we shall later identify with the beginning of stage III work hardening. (The drawn lines in Figs 2, 3 and 4 represent curves calculated on the basis of a phenomenological model that will be proposed further below.)

Figure 3 shows similar data for the single-slip orientation over a wide range of temperatures [27]. The same trend is evident up to about half the melting temperature. For higher temperatures, the lines appear curved from the beginning, although they still go through the origin. We conclude that equation (4) is valid in the limit of small strains at all temperatures, and over a finite range of strains at low temperatures.

Temperature changes can be analyzed in a similar way. Taking account of the temperature dependence of the elastic moduli, and of the fact that temperature changes are usually far from infinitesimal, equation (4) leads to

$$\Delta \frac{\sigma}{\mu} \equiv \frac{\sigma_1}{\mu_1} - \frac{\sigma}{\mu} = \left(\frac{s_1}{s} - 1 \right) \cdot \frac{\sigma}{\mu} \quad (7)$$

where σ_1 and μ_1 refer to measurements at the reference temperature. The term s_1/s is the 'Cottrell-Stokes ratio' [28] for the strain dependent part of the flow stress; when it is constant ('Cottrell-Stokes law'), a plot of $\Delta(\sigma/\mu)$ vs σ/μ should give a straight line as the strain is varied.

Bullen and Hutchison [29] have undertaken a series of temperature change experiments on copper polycrystals. They measured the increase in flow stress (back-extrapolated through the transient) upon

a change to 77 K after various prestrains at various temperatures. Figure 4 shows our reevaluation of their data according to equation (7). The lines are curved, the more so the higher the temperature. This again suggests systematic departures from the Cottrell-Stokes law at larger strains: the earlier and more marked the higher the temperature.

The failure of equation (4) to hold at larger strains can only mean that the factor s decreases with increasing strain. If this were due to a decrease in $\dot{\alpha}$ as a result of the increasing stress [24], the departures should be the same at a given stress for all temperatures. Figures 3 and 4 are evidence against this interpretation. We conclude that the increasing departure from the Cottrell-Stokes law at high strains must be due to a decrease in the rate-dependent flow stress factor $s(\dot{\epsilon}, T)$. In the following, we shall establish a correlation between the departures from the Cottrell-Stokes law and the increasing importance of dynamic recovery, i.e. the decreasing strain-hardening rate. This interpretation unifies the strain and temperature dependencies.

ATHERMAL HARDENING AND DYNAMIC RECOVERY

Strain hardening is due to the evolution of the dislocation density with strain. According to equation (4), it is appropriate to evaluate it as [30-32]

$$\frac{\sigma}{d\epsilon} = \frac{(\alpha ub)^2}{2} \frac{d\rho}{d\epsilon} \quad (8)$$

when α is independent of strain. The evolution rate has been described as being composed of two parts [3, 33, 34]:

$$\frac{d\rho}{d\epsilon} = \frac{1}{\Lambda b} - L_r N_r \frac{v_r}{\dot{\epsilon}} \quad (9)$$

The first term describes the storage rate by a mean free path Λ ; the second a dynamic recovery rate that renders dislocation segments of length L_r ineffective as N_r of them per unit volume rearrange at a rate v_r .

The distinction between a hardening and a dynamic recovery component of the evolution rate becomes meaningful by virtue of the hypothesis that the first is of a strictly geometric or statistical nature, and thus athermal, whereas the second is profoundly influenced by thermal activation. In particular, the mean free path Λ may be assumed to be proportional to the dislocation spacing $1/\sqrt{\rho}$, and thus to $1/\sigma$; then one may write equations (8) and (9) to read

$$\frac{\sigma}{d\epsilon} = \sigma \cdot [\Theta_h - \Theta_r(\dot{\epsilon}, T, \sigma)] \quad (10)$$

where all the constants have been lumped into a 'hardening rate' Θ_h : an athermal contribution to

† The small intercept at positive stresses corresponds to an initial part of the stress strain curve that is concave upwards [31, 32].

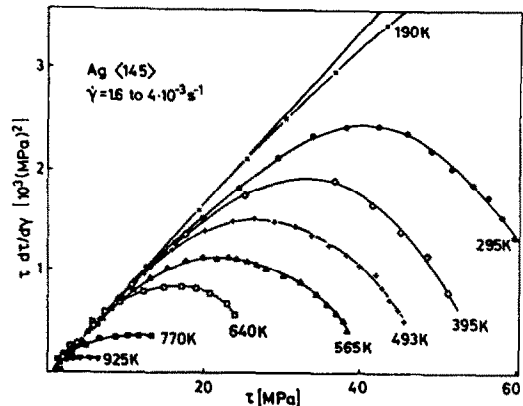


Fig. 5. Strain hardening of silver single crystals in single slip, for a wide range of temperatures [32]. The ordinate is proportional to the rate of evolution of the dislocation density, which is proportional to the flow stress τ in the athermal hardening regime.

$\Theta \equiv d\sigma/d\epsilon$. By analogy, the dynamic recovery term is described as a (negative) contribution to Θ (which is still stress dependent).

Figure 5 shows strain-hardening data for silver single crystals in single slip, over a wide range of temperatures [32]. The evaluation was done according to equation (10), using resolved shear stresses τ and shears γ . It is obvious that the initial behavior is well described by a straight line through the origin,† for all temperatures, confirming the existence of an athermal hardening rate Θ_h . Up to about half the melting temperature, the data follow a straight line for a finite range of flow stresses: this is 'stage II' hardening. The higher temperature data exhibit an initial slope close to the same Θ_h , but are curved from the beginning. This behavior can also be described by equation (10).

For polycrystals and for single crystals in polyslip the behavior is similar. Figure 6 compares these with the single-slip case for silver at room temperature [35]. The $\langle 100 \rangle$ crystal displays a well-defined stage II, i.e. a constant slope over a significant range of flow stresses; for the $\langle 111 \rangle$ orientation and the polycrystal, this range is smaller.

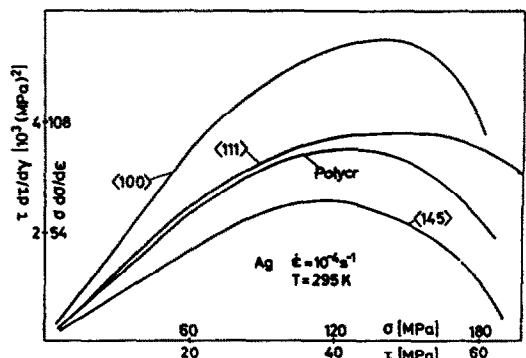


Fig. 6. As Fig. 5, for silver single crystals of different orientations and polycrystals at room temperature [35].

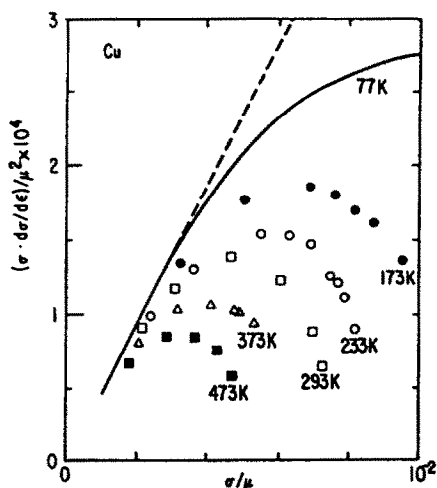


Fig. 7. As Fig. 5 but for copper polycrystals. Data evaluated from Bullen and Hutchison [29].

Copper polycrystals have been investigated over a wide range of temperatures [29] and are plotted according to equation (10) in Fig. 7. The continuous curve at 77 K exhibits stage II behavior over a significant range of stresses. The curves at higher temperatures were available on a discontinuous basis only, and not below about 3% strain. The points in Fig. 7 are, nevertheless, most easily interpreted by ascribing to them an initial behavior that is identical for all of them, and from which deviations occur the earlier the higher the temperature. Thus, all of them can be described by the superposition of an athermal hardening rate θ_h and a temperature sensitive dynamic recovery component θ_r . The value of θ_h used to plot the straight line that characterizes the low-strain asymptote in Fig. 7 corresponds to $\mu/200$ in resolved shear stress and strain (using [10] a Taylor factor of 3.06).

Summarizing the strain hardening data, we assert that an initial athermal hardening rate can be identified for both single crystals and polycrystals at all temperatures, even when a 'stage I' in the sense of an extended straight portion of the stress strain curve cannot. The existence of this athermal hardening rate θ_h , which is always within a factor 2 of the value $\mu/200$ on a resolved basis, allows one to define a dynamic recovery rate θ_r as the deviation of the actual net slope of the stress strain curve from θ_h . The temperature and strain-rate sensitivity of strain hardening reside in the dynamic recovery term.

A number of small corrections must now be applied to the somewhat simplified exposition given above. Firstly, it bears repeating that the temperature dependence of the elastic modulus superimposes on any due to dynamic recovery. Secondly, there is a trivial strain-rate and temperature dependence through the term $s(\dot{\epsilon}, T)$ that enters into α in equation (10) [see equation (4)]. Thus, one may write the strain-hardening rate as

$$\Theta \equiv \frac{d\sigma}{d\epsilon} = s(\dot{\epsilon}, T) \cdot \frac{\mu(T)}{\mu_0} \cdot (\theta_h - \theta_r(\dot{\epsilon}, T)). \quad (11)$$

The temperature dependence of the shear modulus has been taken account of in Fig. 7, that of s has not. The reversible flow-stress change data reviewed in the last section show that the change of s with temperature is small compared to that of θ_r . (One is approaching the flow stress plateau in a σ - T diagram.) In particular, the somewhat faster change of s at larger strains cannot account for the large rate sensitivity of strain-hardening at large strains.

Finally, we have noted in previous sections that α may decrease slightly with increasing deformation; if this is taken into account, equation (8) was not a proper derivative of equation (4), but should be replaced by

$$\frac{\sigma}{\alpha} \frac{d\alpha}{d\epsilon} = \frac{(\alpha\mu b)^2}{2} \frac{d\rho}{d\epsilon} + \sigma^2 \frac{d \ln \alpha}{d\epsilon}. \quad (12)$$

A decrease in α with strain could thus, through both terms on the right-hand side of equation (12), conceivably explain significant deviations from linear strain hardening [36]. For an explanation of the entire stage III, however, this would require α to change by one or more orders of magnitude, depending on temperature, for the data shown in Figs 5-7; this is incompatible with the direct data shown in Fig. 1, and indirectly also with the relatively mild change with strain of the rate and temperature sensitivity (Figs 2-4).

We shall claim below that the cause-and-effect relation is precisely the reverse: the evolution law (rather than the flow stress kinetics described by α) exhibits evidence of a new mechanism (viz. dynamic recovery) becoming more important at larger strains and higher temperatures, and this mechanism in turn influences flow stress kinetics; the feed-back through the variation of α on the evolution rate, according to equation (12), is small.

CORRELATION BETWEEN STRAIN-HARDENING AND FLOW-STRESS KINETICS

Comparison of the strain hardening behavior shown in Figs 5, 6 and 7, with the reversible flow-stress kinetics displayed in Figs 2, 3 and 4, respectively, suggests that there is a correlation between the two: the deviations from Cottrell-Stokes behavior occur at about the same stress at which strain hardening deviates from being athermal, i.e. at the beginning of 'stage III'. Two separate points are worth making by way of abstracting this observation:

1. There is indeed a 'behavior' that obeys all the simplest rules to a good approximation: hardening is athermal [equation (1b)] or even constant ('stage II'), and the strain-dependent flow stress contribution consists of a product of a rate-sensitive term and a strain-dependent term. We will refer to this case as 'Cottrell-Stokes behavior'. It spans a finite regime of (low) strains at low temperatures, but still describes asymptotic initial behavior at higher temperatures. This is true for single crystals and polycrystals.

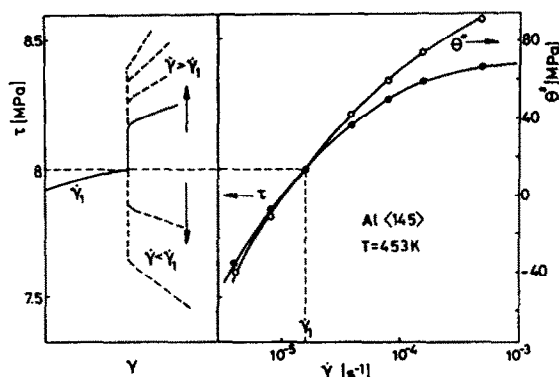


Fig. 8. Left: schematic of effect of strain-rate changes on stress-strain curve. Right: quantitative evaluation of instantaneous flow stress τ and strain-hardening rate Θ as a function of strain rate $\dot{\gamma}$. Data on aluminum at half the melting temperature [38].

2. Deviations from this simple behavior occur increasingly as dynamic recovery becomes more significant: to the degree that strain hardening becomes rate sensitive, the Cottrell-Stokes law breaks down. Strain hardening is the more rate sensitive the lower it is; by extrapolation, this regime includes steady-state flow [34, 37].

We will now demonstrate the correlation between strain-hardening and flow-stress kinetics on one further set of experiments, in which the effects of strain-rate changes on both the flow stress and the hardening rate are measured. They also manifest the need to introduce an additional parameter to characterize the structure or state in this regime.

The dependence of a property on strain rate and temperature is best tested at constant structure or state. This concept was introduced by Cottrell [28] for the flow stress (and used earlier in this paper); it holds as well for the rate of evolution [9]. At a given state (for example, after identical histories), the initial strain-hardening rate in new tests under various conditions may depend on the then current strain rate and temperature just as the flow stress does. Unfortunately, strain-hardening data have usually not been reported upon abrupt changes in strain rate or temperature, and they are not available for the two materials discussed until now.

Figure 8 illustrates data for aluminum single crystals deformed in single slip at half the melting temperature [38] (similar data have been obtained for polycrystalline aluminum alloys [39]). After changes in strain rate of different sign and magnitude, both the flow stress and the initial strain-hardening rate change, and they appear to do so in a correlated fashion. Increases in strain rate lead to increases in both flow stress and strain hardening, at a decreasing rate as the magnitude of the change gets larger. Decreases in strain rate lead to decreases—larger in magnitude—of both the flow stress and the initial hardening rate.

The dependence of the initial hardening rate Θ^* on the new strain rate $\dot{\gamma}$ is shown in Fig. 9 for aluminum single crystals that were prestrained at $T = T_M/2$ to a flow stress of 10 MPa at a fixed $\dot{\gamma}_1$ [38, 40]. It is seen that decreases in strain rate lead to negative initial hardening rates: the well-known phenomenon of work softening [41]. The curve shown in Fig. 9 can be described by

$$\Theta^* = \Theta_h - c \cdot \dot{\gamma}^{-1/n} \quad (13)$$

where c is a constant that should depend on the state, and $n \approx 5$.

The inset in Fig. 9 shows that the *change* in strain-hardening rate upon a *change* in strain rate is larger than the *difference* between the Θ values attained after continuous tests at the two strain rates from the annealed state to the same flow stress. Had the strain rate been *decreased*, the initial value Θ^* can even be negative, whereas the strain-hardening rate measured in continuous tests is always positive.

Thus, there is a *transient* in strain-hardening behavior upon a change in conditions. It lasts for typically about 3–5% strain. This transient constitutes evidence for the necessity of introducing an additional structure parameter [2, 9]. If the flow stress characterized the state completely, then Θ should be the same at the same σ for the same current test conditions of $\dot{\gamma}$ and T . This is not the case; however, after completion of the transient, it becomes true. Thus, the additional state parameter must reach some steady-state value for the given conditions, after a few percent strain.

The 'long transient' discussed here is different in kind from the short 'inverted' transient that usually extends for only a fraction of a percent strain [2]. The long transient is observed only in stage III of work hardening, when dynamic recovery becomes increasingly important. In the athermal hardening regime, the strain-hardening rate is the same before and after a strain-rate change.

This completes our exposition of experimental facts. *We conclude that with increasing importance of*

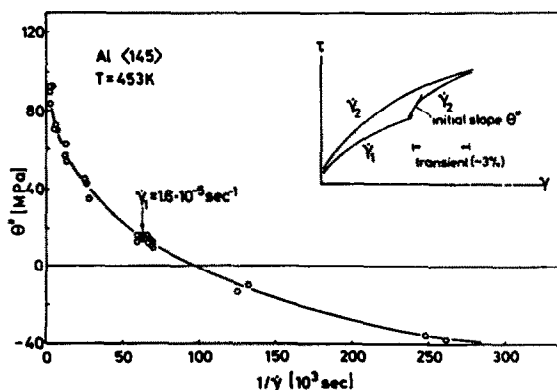


Fig. 9. The initial strain-hardening rate Θ^* after strain-rate changes away from $\dot{\gamma}_1$, which was used in prestraining to 10 MPa [38, 40]. Inset: a transient of about 3% duration follows upon a strain-rate change in stage III.

dynamic recovery, the various simple assumptions that were found to hold true during the initial 'Cottrell-Stokes behavior' break down simultaneously; equations (2a) to (2c) are needed as a general foundation.

A PHENOMENOLOGICAL MODEL

Glide kinetics is generally described by an Arrhenius equation [2, 7]:

$$\dot{\epsilon} = \dot{\epsilon}_0 \exp \left[- \frac{\Delta G(s_A)}{kT} \right] \quad (14)$$

where $\Delta G(s_A)$ is meant to express the stress dependence of the activation free enthalpy. Inversion of equation (14) gives a special form, when equation (14) holds, for the stress factor s introduced in equation (4); namely

$$s \equiv \frac{\sigma}{\hat{\sigma}} = s_A(\dot{\epsilon}, T) = \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right)^{1/m}. \quad (15)$$

In the last part of the equation, we have expressed s_A as a power-law in the strain rate, which is often used for convenience in phenomenological expressions. It is truly equivalent to an inversion of equation (14) only when

$$m \equiv \frac{\partial \ln \dot{\epsilon}}{\partial \ln s_A} \Big|_T = \frac{1}{kT} \cdot \left(- \frac{\partial \Delta G}{\partial \ln \sigma} \Big|_T \right) \quad (15')$$

is independent of stress, as it often is to a sufficient approximation. The temperature dependence of m at constant strain rate [which is more complicated than the explicit temperature dependence evident in equation (15')] causes the temperature dependence of s and thus of the flow stress in the power-law description (15) [7].

Equation (15) is a specific example of Cottrell-Stokes behavior [equation (4)]: the flow stress σ is a product [26] of $\hat{\sigma}$, which depends only on strain, and s , which depends only on strain rate and temperature. This behavior would be violated if, for example, $\dot{\epsilon}_0$ were not constant but depended on $\hat{\sigma}$. A mild dependence of this kind is actually likely [7]; however, it would go in the wrong direction to explain the observed deviations from Cottrell-Stokes behavior at large flow stresses—and could not explain the correlation with dynamic recovery and the appearance of transients.

In order to describe and to test the postulated connection between deviations from Cottrell-Stokes kinetics and dynamic recovery, let us instead replace equation (15) by

$$s = s_A \cdot f(\theta_r) \quad (16)$$

† For the strain-hardening rate, we have used the 'steady-state' value at the base strain rate, θ ; nevertheless, for its rate sensitivity, we have used the instantaneous dependency given by equation (13). Moreover, we have assumed this equation, which had been derived for Al single crystals at half the melting temperature, to hold, in its functional form, for silver mono- and polycrystals at all temperatures.

where the as yet arbitrary function f must go to 1 as θ_r goes to zero, and must decrease as θ_r increases. We shall use two easy heuristic representation of such a function [inserting equation (15) for the general glide kinetics term s_A]:

$$s = \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right)^{1/m} \cdot \exp \left(-F \cdot \frac{\theta_r}{\theta_h} \right) \quad (17a)$$

and

$$s = \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right)^{1/m} \cdot \left(1 - F \cdot \frac{\theta_r}{\theta_h} \right) \quad (17b)$$

where θ_h has been introduced merely for normalization, and F is an adjustable parameter (not necessarily constant, but assumed so in the following); since it always comes out small compared to 1, the two expressions (17a) and (17b) are in fact equivalent, and will be used interchangeably, depending only on convenience.

Equation (17a) was originally derived on the basis of a physical mechanism, which has not as yet been published in detail, but was summarized in Ref. [11]. It is based on the contribution made to glide kinetics by the 'dislocation rearrangement strain rate' during dynamic recovery, and also provides a structural interpretation for a second state parameter [11, 42]. In the present paper, we shall merely use equations (17) as empirical relations and prove that they describe the data presented here quite well.

Equation (17a) leads to the following expression for the rate sensitivity:

$$\frac{\partial \ln s}{\partial \ln \dot{\epsilon}} \Big|_T = \frac{1}{m} - \frac{F}{\theta_h} \frac{\partial \theta_r}{\partial \ln \dot{\epsilon}} \Big|_T \quad (18)$$

and using equation (13) for the instantaneous rate sensitivity of the hardening rate:

$$\frac{\partial \sigma}{\partial \ln \dot{\epsilon}} \Big|_{T, \hat{\sigma}} = \frac{\sigma}{m} \cdot \left(1 + \frac{m}{n} F \frac{\theta_r}{\theta_h} \right). \quad (19)$$

The equation was plotted as a solid line through the data points in Figs. 2 and 3, using θ_r/θ_h as evaluated from Figs. 5 and 6, respectively†. We found that a value of $m/n \cdot F = 1$ fit all the room-temperature curves in Fig. 2; for higher temperatures, the value is given in parenthesis on the respective curves in Fig. 3. (The temperature dependence is consistent with that of m/n , and thus with $F \approx \text{constant}$.) The fit is remarkable. Note in particular that the very different stress strain curves for the different single crystal orientations and the polycrystal of silver give very similar flow stress kinetics.

Temperature changes should similarly give, instead of equation (7), a flow stress change according to

$$\Delta \frac{\sigma}{\mu} \equiv \frac{\sigma_1}{\mu_1} - \frac{\sigma}{\mu} = \left[S - 1 - S \cdot \Delta \left(F \frac{\theta_r}{\theta_h} \right) \right] \cdot \frac{\sigma}{\mu} \quad (20)$$

where the linearized equation (17b) has been used, and S is the initial Cottrell-Stokes ratio at the two

temperatures (s_{A1}/s_A). The copper data in Fig. 4 were fit using equation (20) with $\Delta\theta_r = -\theta_r(T)$, and evaluating θ_r/θ_h for each temperature from Fig. 7 (employing a computer smoothing technique). A single value $F = 1/30$ produced the good fit in Fig. 4.

Equations (17) thus describes these data very well. Note that the correction factor for the flow stress is always near 1 ($F < 0.1$, $\theta_r/\theta_h < 1$); nevertheless, the influence on the rate and temperature sensitivity is profound.

A further ramification of the new flow stress law expressed by equation (16), as inserted into equation (4), is that an Arrhenius equation no longer describes flow kinetics when the rate of dynamic recovery is substantial. The specific form given in equation (17) is such that the observed flow stress should be lower than that predicted by an Arrhenius equation [equations (14) and (15)] when the strain rate is low or the temperature high. This is exactly what is observed in the high-temperature flow-stress drop [2, 5], and during long-term stress relaxation [43, 44]. In all these cases, the curvature in a plot of σ vs T or $\ln \dot{\epsilon}$ has the wrong sign to be consistent with thermally activated glide over a physically reasonable obstacle in which the activation area increases as the stress decreases [44]. Figure 8 of the present paper displayed another case of such an unexpected curvature, but here for the flow stress after a strain-rate change. The temperature was there not quite in the flow-stress-drop regime, but the strain was large enough to produce substantial dynamic recovery. *The critical criterion for the departures from classical behavior is a low strain-hardening rate.*

BASINSKI'S MODEL

In the preceding we have presented a point of view that emphasizes the correlation between the observed breakdown in Cottrell-Stokes behavior with the breakdown in athermal strain hardening. The effect is mainly ascribed to an increase in the rate sensitivity of α in the $\sigma - \sqrt{\rho}$ relationship in equation (4) as the dynamic recovery component θ_r of strain hardening increases. If one expresses the rate sensitivity as

$$\beta \equiv \frac{\partial \sigma}{\partial \ln \dot{\epsilon}} = \frac{\partial \alpha}{\partial \ln \dot{\epsilon}} \cdot \mu b \sqrt{\rho} \quad (21)$$

then $(\partial \alpha / \partial \ln \dot{\epsilon})$ is claimed to be a function of θ_r . There is also a small decrease in the absolute magnitude of α which, however, is negligible in the proposed formalism.

Just the opposite point of view has been taken by Basinski [22, 24] in a previous attempt to explain the deviations from Cottrell-Stokes behavior at large strains. It is based on the assumption that the absolute value of α in equation (4) decreases with increasing strain. This follows from the postulate that the flow stress is proportional to the dislocation line tension, and that the outer cut-off radius in the line tension is

proportional to the dislocation segment length [36]. If the inner cut-off radius is called r_0 , this gives

$$\sigma = \alpha_0 \mu b \sqrt{\rho} \ln(1/r_0 \sqrt{\rho}). \quad (22)$$

The effective value of $\alpha = \alpha_0 \ln(1/r_0 \sqrt{\rho})$ then decreases with increasing dislocation density. On the other hand, it is assumed that the rate sensitivity is not proportional to the line tension, but

$$\beta = B \cdot \mu b \sqrt{\rho} \quad (23)$$

with B independent of ρ . Then, the relationship between flow stress and rate sensitivity can be expressed by

$$\sigma = C \beta \cdot \ln(D/C \beta) \quad (24)$$

where $C \equiv \alpha_0/B$ and $D \equiv \alpha_0 \mu b/r_0$. Equation (24) describes a curve, not a straight line, in a Haasen plot and is used to explain deviation from Cottrell-Stokes behavior.

The above description was based on a model [24] in which the flow stress is composed of two parts: an athermal component described by equation (22), and a thermally activated component which was assumed to be so small compared to the athermal one that it was not even included in equation (22) as an additive term [45], even though it is solely responsible for equation (23). The crucial assumption is that the two components have different ρ -dependencies. In our opinion, this represents a superposition rule which is hard to justify theoretically and certainly refers to a very special obstacle distribution. In particular, it appears to be in conflict with basic features of Cottrell-Stokes behavior.

The Cottrell-Stokes law is generally interpreted as experimental evidence for the athermal and thermal stress components (if indeed they have a separate existence) to be controlled by the same obstacles, namely forest dislocations [46]. The line tension controls the area enclosed by dislocation segments at the critical bow-out, and thus the statistics of the effective obstacle spacing along the gliding dislocations. Whether and in which form it appears in the macroscopic flow stress relation depends on the obstacle strength and on the detailed averaging procedure, and cannot be stated in a general way [7, 47]. In any case, however, it should enter both stress components in precisely the same way when they are controlled by the same obstacles.

If nevertheless the distinction between the ρ -dependence of the two stress components were accepted on a phenomenological basis, it could indeed explain the sign of the observed deviations from Cottrell-Stokes behavior; to explain the observed magnitude, however, the inner cut-off radius would have to have a value of $r_0 \approx 15b$ for the data reported by Basinski [24] (and also for our evaluations at room-temperature, as reported below), which is not easy to rationalize. Furthermore, the strong $\alpha(\rho)$ dependence implied by this value tends to overexplain the possible mild departure from the $\sigma \propto \sqrt{\rho}$ proportionality shown in

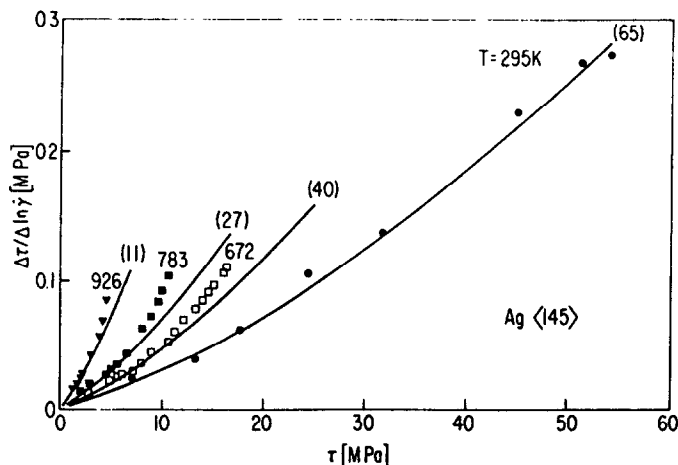


Fig. 10. Data from Fig. 3, but with curves derived according to Basinski's model [23]. The cut-off radius r_0 , which determines the curvature, was matched to the room-temperature data, the other adjustable parameter was matched to the low-stress regime for $T = 672, 783$ and 986 K, and is given in parenthesis.

Fig. 1. (See also the detailed discussion on the data in Fig. 1.)

Finally, and for the present paper most significantly, Basinski's model is based on a modification of the *athermal* part of the flow stress in equation (22). Therefore, in the framework of his approach, a description of $\beta(\sigma)$ plots for different temperatures should be possible according to equation (24) with a constant value of D (and thus of r_0), and an adjustable value of C [and thus of B in equation (23)] for each temperature. This procedure was applied to the data in Fig. 3, and the result is shown in Fig. 10. The value of $D = 340$ MPa was chosen such as to give the best fit for the room-temperature data; it is equivalent to Basinski's value for $r_0 = 15b$, when one assumes $\alpha_0 = 0.2$, which is consistent with the data in Fig. 1. Now the values for C were determined such as to match the low-stress (Cottrell-Stokes) regime for a series of temperatures for which the $\beta(\sigma)$ -data were well documented (Fig. 3), namely $672, 783$ and 986 K; these C -values are given in parenthesis on each curve. As can be seen, the data points deviate considerably from the calculated curves.

A good fit of the data by equation (24) could, in fact, be obtained—but only by allowing the quantity D , i.e. the inner cut-off radius, to vary with temperature. For the data at 783 K, for example, this required (for $\alpha_0 = \text{constant}$) an increase in r_0 by a factor of 20 compared to room temperature, i.e. an increase from $r_0 = 15b$ to $300b$.

We conclude that Basinski's model, even if it could be theoretically founded in a more satisfactory way, cannot explain the large-strain departures from Cottrell-Stokes behavior at higher temperatures. In light of this rather limited success, it should not be used as independent support for systematic departures from the $\sigma \propto \sqrt{\rho}$ proportionality either.

SUMMARY

We assert that the experimental data for single crystals of various orientations and for polycrystals of pure f.c.c. metals at temperatures up to at least half the melting point have the following cardinal features:

1. An athermal hardening rate exists, which dominates at low strains and low temperatures ('stage II'), but describes initial asymptotic behavior at all temperatures. Departures from this behavior at larger strains ('stage III') are strongly rate and temperature sensitive.
2. The Cottrell-Stokes law is obeyed (excluding the easy-glide regime from our considerations) over precisely the same range, or in the same limit, in which strain hardening is athermal.
3. The Arrhenius equation, which gives an adequate description of flow kinetics in the athermal hardening regime, also breaks down when the rate of dynamic recovery becomes substantial.
4. The instantaneous response of the strain-hardening rate to strain-rate changes is correlated with that of the flow stress, in the dynamic recovery regime. The classical decoupling of 'reversible' flow stress kinetics 'at constant structure' from the evolution kinetics is no longer possible in this regime.
5. A nontrivial dependence of flow stress kinetics on evolution kinetics demands the introduction of a second structure/state parameter; this is in agreement with the appearance of an additional transient in just the same regime.
6. The proportionality between the flow stress and the square-root of the dislocation density is well documented for copper at room temperature. Mild deviations are probably chiefly due to systematic differences between etch-pit and TEM results.

These observations were quantitatively well described on the basis of a phenomenological model, in which the classical equation for glide kinetics is modified by a factor that decreases from 1 as the rate of dynamic recovery increases. Thus, all the observed departures from classical behavior are linked in their importance to the smallness of the strain-hardening rate - whether this be at large strains or at high temperatures.

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